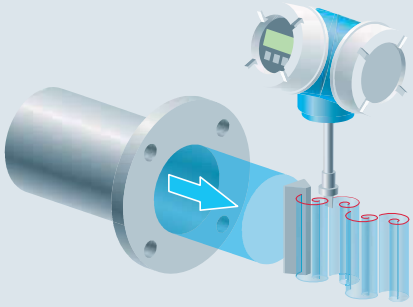
Vortex flowmeters

Proline Prowirl – Robust and universally applicable. For measuring the volume and mass flow of liquids, gases and steam up to 250 bar (3626 psi) and 450 °C (842 °F) reliably.



Measuring principle

This measuring principle is based on the fact that turbulence forms downstream of obstacles in the flow, such as a bridge pier.
Inside each vortex flowmeter, a bluff body is located in the middle of the pipe. As soon as the flow velocity reaches a certain value, vortices form behind this bluff body are detached from the flow and transported downstream. The frequency of vortex shedding is directly proportional to the mean flow velocity and thus to the volume flow.
The detached vortices on both sides of the bluff body generate alternately a local positive or negative pressure that is detected by the capacitive sensor and fed to the electronics as a primary digital, linear signal.

Advantages at a glance

- Universally suitable for measuring liquids, gases and steam
- Largely unaffected by changes in pressure, density, temperature and viscosity
- High long-term stability: no zero-point drift, lifetime K-factor
- No moving parts
- Little pressure loss
- Easy installation and commissioning
- Large turndown of typically 10:1 to 30:1 for gas/steam, or up to 40:1 for liquids
- Wide temperature range: –200 to +400 °C (–328 to +752 °F) (450 °C / 842 °F on demand)

Measuring principle movie:
www.eh.digital/2XzvvJ1

In chemicals, petrochemicals, power engineering and heat supply, a wide variety of fluids can be measured using vortex flowmeters, e.g. saturated steam, superheated steam, compressed air, nitrogen, liquefied gases, flue gases, carbon dioxide, fully demineralized water, solvents, heat-transfer oils, boiler feedwater or condensate. Vortex flowmeters are also in widespread use for measuring mass flow. Therefore, modern vortex meters such as the multivariable Prowirl 200 are built for more than merely measuring volume flow, and come complete with pressure and temperature sensors as well as a flow computer.

Whenever gas mass flow has to be measured, external pressure values can be read in digitally and with high accuracy via HART, PROFIBUS or FOUNDATION Fieldbus. Prowirl 200 is also available with reduced line sizes, which permits measurements even at very low flow velocities – with the same installation length and accuracy.

Prowirl 200 is the world's first vortex flowmeter with the option of monitoring the steam quality and immediately generating an alarm message in case of wet steam. Prowirl can also be used for flow monitoring systems up to SIL 2 and SIL 3 and is independently evaluated and certified by TÜV Rheinland in accordance with IEC 61508.





Prowirl sensors

Prowirl D

Compact wafer device



- With centering rings for high fitting accuracy
- Worldwide standardized installation length (65 mm) enables one-to-one replacement of orifice plates
- Sensor made of stainless steel (CF3M)
- PN 10 to 40 (Class 150 to 300), 10 to 20K
- –200 to +400 °C (–328 to +752 °F)
- DN 15 to 150 (½ to 6")

F **L** E X

Prowirl F

Versatile standard device



- Suitable for detecting wet steam
- Correction function for short inlet runs
- Worldwide standardized installation lengths
- Sensor made of stainless steel (CF3M/316/316L) or Alloy C22
- PN 10 to 100 (Class 150 to 600)
- –200 to +400 °C (–328 to +752 °F) (450 °C / 842 °F optional)
- DN 15 to 300 (½ to 12")

F L **E** X

Prowirl R

For low flows



- With a single and double line size reduction for:
  – Increasing the flow velocity
  – Extending the lower measuring range
- PN 10 to 40 (Class 150 to 300), 10 to 20K
- DN 25 to 200 (1 to 8") (single reduction)
- DN 40 to 250 (1½ to 10") (double reduction)

F L **E** X

Prowirl O

The high-pressure specialist



- Flange or butt-weld version
- Sensor made of stainless steel
- PN 160 to 250 (Class 900 to 1500)
- –200 to +400 °C (–328 to +752 °F) (450 °C / 842 °F optional)
- DN 15 to 300 (½ to 12")

F L **E** X

Integrated pressure measurement

As a multivariable vortex flowmeter, Prowirl 200 provides an all-in-one solution: simultaneous measurement of mass flow, corrected volume flow, energy flow, temperature and, for the first time, even the process pressure.
Regardless of the amount of fluctuation in the process variables, Prowirl makes it possible to measure with high precision and thus offers extensive energy management for different fluids like steam, gases, water or hydrocarbons.



Robust DSC sensor

Endress+Hauser's unique, patented DSC (Differential Switched Capacitor) sensor ensures high-precision measured values even under the toughest conditions and features a lifetime calibration factor. With an installed base of over 500 000 devices, this sensor concept has been proving its value for decades.
The sensor is highly resistant to:

- Vibration
- Water hammer and condensation-induced water hammer
- Dirty fluids
- Temperature shocks (>150 K/s)

Optionally, Prowirl is also available with an integrated pressure and/or temperature sensor, for example for direct mass measurement of wet steam, saturated steam and superheated steam.



Proline transmitters

Proline 200



Two-wire loop-powered technology (4–20 mA)
- Four-line display with push buttons or touch control
- Display module with backup and transfer function for configuration data
- HART, PROFIBUS PA, FOUNDATION Fieldbus and PROFINET-APL with pulse/ frequency/switch output
- Heartbeat Technology for diagnostics, verification and monitoring
- Versatile system integration:
  - Current input for reading in external measured variables such as pressure or temperature (optional)
  - Current output for multiple measurement parameters (optional)

Advantages of the Proline 200 transmitter ▸ page 9 to 11
Prowirl is the world's first vortex flowmeter developed entirely in accordance with IEC 61508, allowing it to be used in SIL 2/3 applications at any time.

Multivariable measurement

Energy management made easy

All industries have utilities with steam, cooling water or hot water. Generating, transporting and distributing these fluids consumes a lot of energy. Therefore, Proline Prowirl 200 offers everything in one device for a comprehensive energy management:
- Integrated flow computer for calculating:
  - Mass, heat and energy flow of steam and liquids
  - Corrected volume flow and energy flows of gases
- Reading in of external temperature and pressure values via HART, PROFIBUS PA and FOUNDATION Fieldbus as well as via an optional current input
- Integrated pressure and temperature measurement for direct mass measurement of saturated steam and liquids (temperature compensation)

For energy management we offer everything from a single source: flow computers, pressure and temperature sensors as well as Data Manager Memograph M RSG45 with software solutions for energy monitoring.



EngyCal RS33 RSG45 Cerabar TM131



One-of-a-kind wet steam detection

Process reliability and efficiency

Many industries use large quantities of steam, the generating costs of which are extremely high. Moreover, the transfer of heat energy is energetically efficient only for saturated steam. Often, however, wet steam is what predominates, since fluctuations in pressure and temperature cause water to condense out, or water gets into the steam lines due to disruptions in the boiler system. The consequences are usually serious:
- Low efficiency for the transmission of energy
- Hazardous water hammer and condensate-induced water hammer
- Heavy corrosion from the salts dissolved in the water carried over

Movie – Steam quality:
Prowirl measures all types of steam right in the pipe inclusive steam quality and condensate quantity.
www.eh.digital/3wFpVSr